

TAPPAHANNOCK-ESSEX COUNTY AP, VA, USA

WMO: 720501

Lat: 37.850N

Lon: 76.883W

Elev: 135

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.6	(c) 19.4	(d) -8.1	(e) 3.6	(f) 24.7	(g) -1.0	(h) 5.2	(i) 29.4	(j) 19.0	(k) 43.5	(l) 16.9	(m) 43.4	(n) 3.2	(o) 300	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 95.2	(d) 76.2	(e) 91.9	(f) 74.7	(g) 90.1	(h) 74.0	(i) 80.1	(j) 87.9	(k) 78.7	(l) 86.0	(m) 77.5	(n) 84.5	(o) 5.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	149.1	82.5	77.0	141.4	81.4	75.4	133.9	80.0	43.7	88.0	42.3	85.8	40.9	84.4	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.1	(b) 13.2	(c) 11.7	(e) 4.8	(f) 98.9	(g) 7.6	(h) 3.1	(i) -0.7	(j) 101.1	(k) -5.2	(l) 103.0	(m) -9.5	(n) 104.7	(o) -15.0	(p) 107.0		
(4) 16.1	(5) 13.2	(6) 11.7	(7) 4.8	(8) 98.9	(9) 7.6	(10) 3.1	(11) -0.7	(12) 101.1	(13) -5.2	(14) 103.0	(15) -9.5	(16) 104.7	(17) -15.0	(18) 107.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	36.6	40.8	47.3	58.2	67.5	75.2	78.9	76.6	71.4	60.0	48.5	42.4	
	DBStd	16.75	10.24	10.73	10.13	9.00	7.83	5.76	4.92	4.54	6.56	8.37	8.61	9.36	
	HDD50	1349	428	299	179	23	0	0	0	0	0	14	128	278	
	HDD65	3847	879	680	554	244	70	3	0	0	17	200	499	701	
	CDD50	4530	14	40	95	269	543	755	897	825	641	325	83	43	
	CDD65	1556	0	1	5	41	148	309	432	361	208	46	3	2	
	CDH74	14342	0	9	56	430	1267	2925	4425	3199	1667	352	11	1	
	CDH80	5779	0	0	4	119	425	1220	2067	1261	594	87	2	0	
Wind	WSAvg	4.3	4.9	5.3	5.8	5.8	4.5	3.9	3.3	2.8	3.3	3.9	4.3	4.3	
Precipitation	PrecAvg	44.0	3.2	2.8	4.2	3.1	3.9	3.6	4.7	4.4	4.4	3.6	3.1	3.4	
	PrecMax	62.3	6.5	6.6	8.7	5.6	9.1	6.3	8.9	13.4	12.7	6.9	9.3	8.3	
	PrecMin	32.7	1.2	0.8	0.2	1.3	1.0	1.3	1.3	1.4	0.8	0.0	0.2	1.0	
	PrecStd	7.0	1.3	1.6	1.8	1.2	2.0	1.4	2.2	2.7	3.1	2.1	2.0	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.0	75.0	79.5	88.4	91.1	98.7	100.0	95.3	94.0	86.8	74.6	72.5	
		MCWB	60.8	61.8	62.7	67.2	71.4	76.8	76.7	77.6	73.1	71.3	63.8	65.1	
	2%	DB	63.3	69.2	74.6	83.6	88.2	92.8	96.8	91.7	90.4	82.0	70.9	68.1	
		MCWB	55.9	57.9	59.5	64.3	70.4	74.0	75.9	76.4	73.3	69.9	61.4	63.6	
	5%	DB	58.8	63.6	69.9	79.1	84.9	90.1	93.0	89.8	86.5	77.6	67.6	63.5	
		MCWB	52.5	54.3	57.4	63.0	70.8	73.2	75.4	75.8	73.5	67.7	59.9	59.6	
	10%	DB	52.7	58.6	64.4	74.7	81.5	87.3	90.3	87.2	82.7	73.4	63.4	58.7	
		MCWB	45.9	50.7	54.9	61.8	68.9	72.5	74.8	74.6	71.2	64.5	57.0	54.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	66.2	66.3	72.7	77.4	79.9	83.5	81.0	79.4	76.6	68.4	69.2
			MCDB	67.0	72.2	72.5	78.6	84.3	91.1	91.0	88.2	85.5	80.3	71.6	70.2
2%		WB	58.7	61.2	63.3	69.0	75.1	77.8	80.9	79.6	77.7	73.7	65.2	63.8	
		MCDB	62.0	65.8	70.1	77.1	82.6	86.9	89.0	86.4	83.0	78.1	68.8	65.9	
5%		WB	52.9	56.3	60.4	66.2	73.1	76.3	78.9	78.4	76.2	70.5	61.2	60.4	
		MCDB	58.2	62.4	67.0	75.1	79.7	84.8	86.6	85.0	80.9	74.9	65.0	63.0	
10%		WB	47.8	50.8	56.5	63.7	71.1	75.1	77.4	77.3	74.6	67.6	58.0	54.9	
		MCDB	51.9	56.9	63.2	71.7	77.3	83.0	85.2	83.6	79.4	72.5	62.8	57.6	
Mean Daily Temperature Range		5% DB	MDBR	17.8	19.5	20.3	22.8	20.5	20.4	19.9	19.1	19.5	20.9	20.2	17.7
			MCDBR	25.4	25.9	27.1	26.2	23.3	23.4	23.0	21.4	22.6	23.6	24.0	21.2
	MCWBR		20.4	16.9	15.3	11.8	9.9	8.7	7.6	8.0	8.8	12.3	15.1	17.6	
	5% WB	MCDBR	23.5	24.2	22.4	21.9	19.3	20.7	19.9	18.3	16.1	18.8	18.9	20.8	
		MCWBR	21.4	17.9	15.5	12.8	9.5	8.9	8.2	8.3	8.0	11.8	14.9	18.9	
Clear-Sky Solar Irradiance	taub	0.304	0.316	0.348	0.390	0.428	0.482	0.509	0.506	0.409	0.354	0.321	0.307		
	taud	2.526	2.480	2.429	2.325	2.252	2.144	2.088	2.081	2.334	2.491	2.541	2.551		
	Ebn at Noon	281	291	290	282	272	256	248	246	266	273	272	272		
	Edh at Noon	24	29	34	40	43	49	51	50	37	29	24	23		
All-Sky Solar Radiation	RadAvg	715	964	1283	1616	1836	2003	1924	1732	1406	1089	809	631		
	RadStd	58	105	103	119	183	143	95	87	141	125	68	54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (77 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+153

Nomenclature: See separate page